

Methods: instrument, assumptions, and
companion track

Synthetic deterministic track: seeded panels, blind estimates, oracle scoring I

The first methodological track is a seeded synthetic instrument. It generates known populations and corpora, hides the answer key from the `ntqr` estimator, and then scores the returned logically consistent evaluations against the supervised oracle that is available only because the data are synthetic.

Synthetic deterministic track: seeded panels, blind estimates, oracle scoring II

Pipeline: panel formation precedes no-answer-key estimation

The instrument runs strictly upstream-to-downstream and is deterministic end to end (every stochastic step is seeded with `numpy.random.default_rng`; figures use `MPLBACKEND=Agg`). One trial is:

1. **Generate** a synthetic expert population of known properties.
2. **Form** a panel from that population with one of four strategies.
3. **Judge** a corpus of items whose ground-truth labels are *known to us but hidden from the estimator*.
4. **Evaluate** the panel with the `ntqr` package *without the answer key*.
5. **Score** the unsupervised estimate against the supervised **oracle** computed from the known labels.

[@fig:pipeline] shows this upstream-to-downstream flow at a glance.

The instrument measures upstream panel formation, not the estimator

Real-Ollama reviewer-panel track: single-model live companion I

The second methodological track uses one live local language model through Ollama: `gemma3:4b`. It is deliberately **single-model** and deliberately not a model-family comparison. The empirical question is whether the same sampling mechanisms studied analytically — representative sortition, random selection, same-bias bloc selection, and expertise-threshold selection — remain visible when one real local LLM is prompted as different postdoctoral-review panelists.

Real-Ollama reviewer-panel track: single-model live companion II

The live track is therefore closer to an instrumented LLM-judge stress test than to an ethnography of human review. LLM-as-judge work has shown that prompted model judgments can be useful but also vulnerable to evaluator-specific artifacts such as position, verbosity, and self-enhancement biases Zheng et al. (2023); systematic position-bias tests likewise show that judge outputs can change with answer order rather than only answer quality Shi et al. (2025). Broader language-model risk work likewise warns against treating fluent model output as an unmediated measurement of social reality Bender et al. (2021). We use one model, bounded decoding, serialized provenance, and synthetic applicant/reviewer metadata precisely so the empirical surface remains auditable and does not masquerade as human-review validation. The two tracks are kept strictly separate and their evidence is never pooled: the synthetic deterministic track remains the controlled spine, while the live track is a companion stress test with separate artifacts, a separate config hash (5161ffe474b3),